



ENGINEERED SOLUTIONS
FOR A CLEANER, SAFER TOMORROW

RELIABLE SOLUTIONS.
REAL PROGRESS.

DOSING
INJECTION
METERING
TRANSFER
CONTROL

RELIABLE
FLEXIBLE
CUSTOMIZABLE

DIAPHRAGM DOSING SKID SYSTEM LNS-DZ-PDD

Multi-Industry Precision Chemical Dosing System



PEOPLE
INDUSTRY
CLEANER
TOMORROW



WATER
TREATMENT



CHEMICAL
PROCESSING



PETROCHEMICAL



UTILITIES
& OTHERS



INDUSTRIAL
PROCESS

SOLUTIONS
FOR A SAFER TOMORROW

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PRODUCT OVERVIEW & APPLICATION SCOPE

1. PRODUCT DESCRIPTION

The **LINESIG LNS-DZ-PDD** is a modular positive-displacement diaphragm metering skid engineered for accurate, repeatable and controllable chemical dosing. The diaphragm isolates the process liquid from the drive mechanism, making this family particularly suitable for corrosive, toxic, hazardous, crystallizing or otherwise difficult chemicals where leakage control and precise metering are important.

The skid can be configured as a single dosing train, duty/standby package or multi-pump assembly. Capacity can be controlled by stroke length, stroke frequency, motor speed or a combination of these methods depending on the selected pump technology. The package may incorporate calibration, pulsation control, back-pressure protection and full PLC/HMI integration.

2. INDUSTRIES

- Water and wastewater treatment
- Oil & gas, refinery and petrochemical facilities
- Fuel terminals and hydrocarbon additive systems
- Power generation and boiler-water treatment
- Chemical processing and specialty chemical plants
- Food, pharmaceutical and process industries where compatible
- Utilities, cooling-water and environmental treatment systems

3. TYPICAL MEDIA

Media Group	Typical Examples	Engineering Consideration
Acids / Alkalis	HCl, H ₂ SO ₄ , NaOH and project-specific solutions	Final wetted materials and diaphragm elastomer selected by concentration and temperature.
Water-Treatment Chemicals	Hypochlorite, coagulants, antiscalants, corrosion inhibitors, biocides	Degassing, crystallization and off-gassing behavior to be reviewed.
Hydrocarbon / Fuel Additives	Methanol, selected additives, dyes, markers, treatment chemicals	Hazardous-area and seal/material compatibility required.
Specialty Chemicals	Catalyst solutions, inhibitors, neutralizing agents, process reagents	Accuracy and low-flow stability often drive pump selection.

TYPICAL DIAPHRAGM DOSING SKID CONFIGURATION

4. SYSTEM ARRANGEMENT

A standard LNS-DZ-PDD package normally includes a chemical day tank or suction connection, suction isolation and strainer as required, diaphragm metering pump(s), pulsation dampener, calibration column, pressure protection, discharge isolation, injection/control valve, instrumentation and a local or remote control system.

Subsystem	Typical Arrangement
Pump Configuration	1 x 100%, 1W+1S, 2W+1S or multi-line arrangement
Suction	Common or individual suction manifold with isolation
Calibration	Graduated calibration column or volumetric verification arrangement
Pulsation Control	Suction and/or discharge pulsation dampener where required
Pressure Protection	Relief valve and/or back-pressure valve
Discharge Control	Injection valve, check valve, isolation valve, optional FCV
Instrumentation	Flow, pressure, level and local indication as required
Control	Manual, local automatic, PLC/DCS ratio or batch dosing

5. DIAPHRAGM PUMP TYPES

- Mechanically actuated diaphragm metering pumps
- Hydraulically actuated diaphragm metering pumps
- Double-diaphragm leak-detection designs for hazardous or critical service
- Solenoid diaphragm pumps for smaller packaged duties
- Variable-speed or stroke-controlled motor-driven diaphragm pumps

TECHNICAL SPECIFICATIONS

6. GENERAL DESIGN DATA

Parameter	Typical / Available Basis
Service	Chemical dosing, treatment chemical feed, additive injection and controlled metering
Flow Range	Project specific; from low-flow dosing to approximately 1.5 m ³ /h per pump in typical package duties
Discharge Pressure	Project specific; commonly 4–16 bar, with higher-pressure designs available
Pump Type	Positive-displacement diaphragm metering pump
Accuracy	Pump dependent; high repeatability when properly calibrated and operated within turndown range
Turndown	Project/pump specific; enhanced by stroke and/or speed control
Operating Temperature	According to diaphragm, valve and wetted material selection
Wetted Materials	PP, PVDF, PTFE, SS316/316L, duplex or alloy as required
Diaphragm	PTFE-faced or other chemically compatible elastomer/composite
Skid Frame	Coated carbon steel; SS304/SS316 optional
Installation	Indoor/outdoor; hazardous-area configuration optional

7. METERING PERFORMANCE

Final pump sizing shall be based on minimum, normal and maximum required chemical flow, discharge pressure, suction conditions, fluid density, viscosity, vapor pressure, gas content and required turndown. The selected operating range should avoid excessively low stroke settings where metering accuracy and valve performance may deteriorate.

CALIBRATION, PULSATION & PRESSURE

8. CALIBRATION SYSTEM

A calibration column may be installed on the suction side to verify actual pump delivery. The column is isolated from the normal supply and used over a timed interval to compare measured volume displacement against the pump setting. The arrangement should include safe venting, drain and isolation provisions.

9. PULSATION CONTROL

Because diaphragm metering pumps produce cyclic displacement, pulsation dampeners may be installed on suction and/or discharge lines to reduce pressure fluctuation, improve flowmeter stability, minimize piping vibration and protect downstream equipment.

10. BACK-PRESSURE & RELIEF

Device	Function
Back-Pressure Valve	Maintains stable discharge pressure and improves dosing accuracy where downstream pressure is low or variable.
Pressure Relief Valve	Protects pump and discharge piping from blocked-line overpressure.
Injection / Check Valve	Prevents reverse process flow and provides controlled chemical entry at the injection point.
Pressure Gauge / Transmitter	Monitors discharge condition and supports alarm/interlock logic.

PIPING, VALVES & MATERIALS

11. PIPING DESIGN

- Keep suction lines short, adequately sized and free of unnecessary high points.
- Avoid excessive suction lift; flooded suction is preferred where possible.
- Provide degassing or vent arrangements for chemicals that release gas.
- Install pulsation dampeners close to the pump connection when required.
- Provide flushing, draining and containment provisions for hazardous or corrosive chemicals.
- Use independent discharge check/injection valves where multiple pumps feed a common process.
- Support piping independently to minimize loads on pump nozzles.

12. MATERIAL SELECTION

Chemical Duty	Typical Material Options
Strong Acids	PVDF, PTFE-lined, selected alloys; exact selection by concentration/temperature
Caustic / Alkali	PP, PVDF, SS316L or compatible material
Hypochlorite / Oxidizers	Special non-metallic / alloy combinations; avoid incompatible elastomers/metals
Hydrocarbon Additives	SS316L and compatible elastomers commonly used
General Water Chemicals	PP, PVC/PVDF, SS316L depending on chemistry

CONTROL, INSTRUMENTATION & AUTOMATION

13. CONTROL MODES

Control Mode	Typical Implementation
Manual	Local stroke / speed setting
Local Automatic	Flow or process-variable control through local controller/PLC
Ratio Dosing	Chemical flow proportional to main process flow
Batch / Totalized	Preset quantity with automatic stop
Remote DCS / SCADA	Setpoint command, permissive, run status and alarm feedback

14. INSTRUMENTATION

- Calibration column or flow-verification device
- Pressure gauge and/or pressure transmitter
- Optional flowmeter on each dosing line
- Chemical tank level transmitter / low-level switch
- Pulsation dampener pressure indication where required
- Leak detection for double-diaphragm pumps
- Local pressure relief and back-pressure devices
- Injection-point non-return valve and isolation

15. AUTOMATION

The package may be supplied with a local PLC/HMI or integrated with the plant DCS/SCADA. Typical logic includes duty/standby selection, automatic changeover, low-level inhibit, high-pressure trip, low-flow alarm, diaphragm leak alarm, motor trip and ESD interface.

SAFETY & HAZARDOUS SERVICE

16. SAFETY FEATURES

Protection	Typical Function
Low Chemical Level	Stops pump before loss of prime or dry operation
High Discharge Pressure	Trips or alarms on blocked injection line
Relief Valve	Mechanical protection of pump and piping
Leak Detection	Detects diaphragm failure in double-diaphragm designs
Bund / Drip Tray	Contains minor leakage and supports drain collection
Emergency Stop	Local E-stop; remote ESD interface optional
Earthing / Bonding	Applied where required for flammable chemical service

17. HAZARDOUS AREA EXECUTION

For flammable solvents, alcohols, hydrocarbon additives or other hazardous chemicals, motors, instruments, solenoids, junction boxes and control equipment shall be selected to match the project area classification. ATEX/IECEx or other hazardous-area certification is project specific and shall be defined during engineering.

18. CHEMICAL HANDLING

Chemical compatibility shall be verified for the pump head, diaphragm, check-valve balls/seats, seals, tubing, piping and calibration equipment. Toxic or highly aggressive duties may require closed venting, double containment, leak detection and dedicated drain/neutralization connections.

APPLICATION CONFIGURATIONS & FAMILY SELECTION

19. TYPICAL CONFIGURATIONS

Configuration	Use
1 x 100%	Single non-critical dosing point
1W + 1S	Critical continuous chemical feed
2W + 1S	Two active dosing lines with common standby
Multiple Independent Trains	Multi-point plant chemical feed
Package with Day Tank	Complete localized chemical storage and dosing package
Remote Supply Manifold	Skid supplied from bulk chemical storage

20. WHEN TO SELECT LNS-DZ-PDD

- Accurate low- to medium-flow metering is required.
- The chemical is corrosive, hazardous or difficult to seal.
- Leak-free separation between process liquid and drive mechanism is preferred.
- Process pressure is above the practical range of a centrifugal dosing pump.
- Ratio control, batch dosing or high turndown is required.
- Pulsating flow is acceptable or can be managed with dampeners.

21. LINESIG DOSING FAMILY

Requirement	Recommended Model
Medium/high-flow smooth blending	LNS-DZ-CEN - Centrifugal
Accurate corrosive chemical dosing	LNS-DZ-PDD - Diaphragm
High-pressure injection	LNS-DZ-PDP - Piston/Plunger
Very low-flow precision additive dosing	LNS-DZ-PDG - Precision Micro Gear

ORDERING DATA & STANDARD SCOPE

22. MINIMUM DATA REQUIRED FOR FINAL SIZING

- Chemical name, concentration and safety classification
- Minimum / normal / maximum dosing rate
- Required discharge pressure and process backpressure
- Suction source, minimum liquid level and suction pressure
- Density and viscosity at operating temperature
- Temperature range and vapor pressure / gas-release tendency
- Required accuracy and turndown
- Duty / standby philosophy and number of injection points
- Required wetted materials and diaphragm material
- Hazardous-area classification
- Control mode and PLC/DCS communication requirement
- Power supply, motor specification and applicable project standards

23. TYPICAL STANDARD SCOPE

Item	Typical Supply
A	Skid structural frame and supports
B	Diaphragm metering pump(s) and motor/drive
C	Suction manifold / isolation arrangement
D	Calibration column
E	Pulsation dampener(s) where required
F	Relief and back-pressure valves
G	Discharge isolation and injection/check valves
H	Pressure instrumentation and optional flowmeter
I	Chemical tank/day tank when specified
J	Local control panel / PLC/HMI when specified
K	Internal piping, tubing and cable supports
L	Documentation: datasheet, GA, P&ID, I/O list, control philosophy and test records as applicable



PRECISE DOSING.

SAFE CHEMICAL HANDLING.

RELIABLE PROCESS CONTROL.

LINESIG LNS-DZ-PD — Engineered for dependable multi-industry chemical metering and injection.